

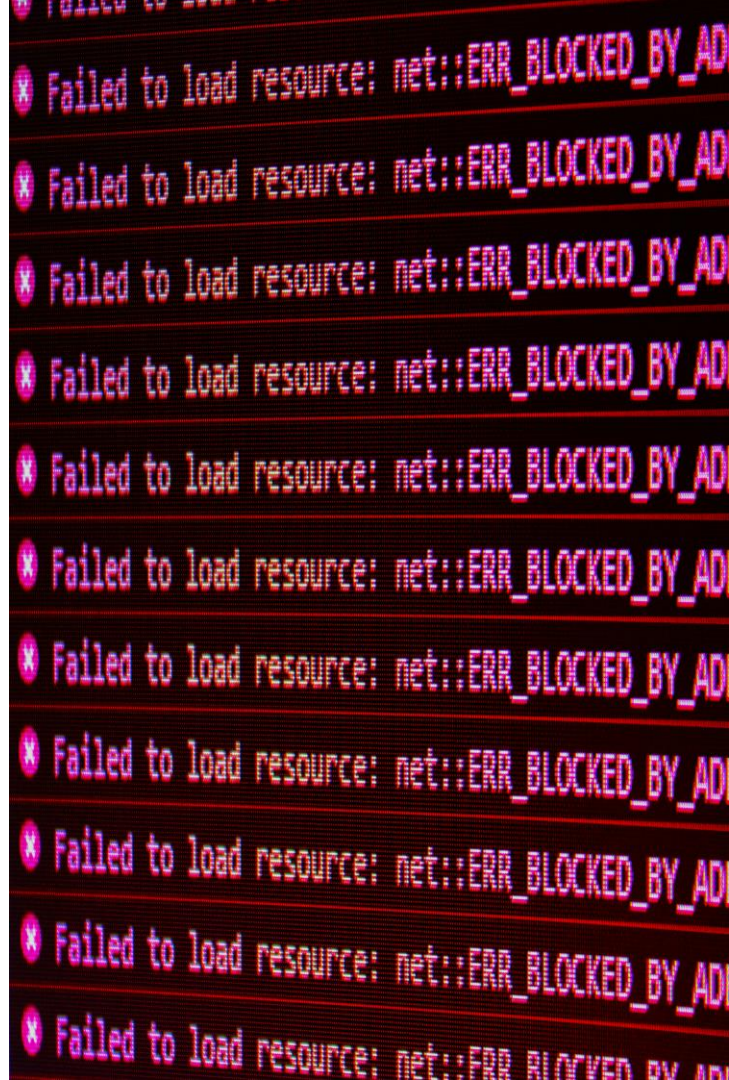
IM-05 — SSL Certificate Expiry Watcher

This briefing presents a professional overview of an AI-powered system that automatically monitors public SSL/TLS certificates, prioritises renewal risk, drafts renewal tasks with an LLM, and delivers alerts and exportable reports for DevOps and SRE teams.



Problem Statement

Organisations manage dozens to thousands of public domains. Expired or misconfigured certificates cause downtime, browser warnings, lost user trust and revenue impact. Manual tracking is error prone and resource intensive — we need automation that detects risk early and prescribes remediation.





Objectives

Automate Monitoring

Periodic scanning of public hostnames from domains.csv to fetch certificate metadata and compute days-to-expiry.

Risk Detection

Classify certificates as HEALTHY, WARNING, or BROKEN and rank by urgency.

AI Task Drafting

Use Ollama + Llama3 to produce step-by-step renewal tasks with an owner placeholder for tickets.

Reporting & Alerts

Export CSV and email-ready Markdown; send critical alerts to Discord; show live status on a Flask dashboard.

System Architecture



Data flows from a source CSV into the SSL Scanner. Engine logic applies status rules and priority, then forwards items to the LLM task generator. Results are persisted to SQLite, surfaced in a Flask dashboard, exported to CSV/Markdown, and critical items trigger Discord alerts.

- Input: domains.csv
- Compute: scanner.py → engine.py
- AI: llm.py (Ollama + Llama3)
- Storage: SQLite
- UI / Exports: Flask, CSV, Markdown
- Alerts: Discord webhooks

Key Modules & Responsibilities



scanner.py

Connects to hostnames, retrieves certificate chain and metadata, computes expiry days. Sample output: {"domain": "google.com", "days_left": 62}.



engine.py

Applies rules (≥ 30 HEALTHY, 10–29 WARNING, < 10 BROKEN), computes incident priority and recommended actions.



llm.py

Ollama + Llama3 generate concise renewal tasks: priority, owner placeholder, and action steps (CSR, certificate upload, service restart).



Storage & Exports

SQLite stores domain, expiry, status, AI task and timestamp. Exports: CSV for audits and Markdown for email-ready ticket content.

Flask Dashboard — UX Highlights

Real-time monitoring with auto-refresh, search and filters. Visual elements include summary cards (Healthy / Warning / Broken / Total), urgency-ranked table and expandable AI tasks per domain.

- Searchable domain list
- Filters: status, priority, owner placeholder
- Ranking: lowest days_left → top of list
- Export buttons: CSV, Markdown



Alerts & Reporting

Discord Alerts

Critical and WARNING items trigger a webhook payload with domain, status and days_left. Example message:

🔒 SSL ALERT 🔒 Domain: wikipedia.org Status: WARNING Days Left: 27. Includes suggested action and link to dashboard.

Exports

CSV: full table for Excel and SIEM ingestion. Markdown: email-ready ticket content with AI-drafted steps and owner placeholder for copy-paste into issue trackers or support emails.



Roadmap



Phase 1 — Core MVP

Scanner, engine rules, LLM task draft, SQLite, Flask dashboard, Discord alerts, CSV/Markdown exports.

Phase 2 — Integrations

Email integration, Jira ticket automation, Docker images, Kubernetes support.



Phase 3 — Observability

Grafana dashboards, SSL trend analytics, multi-user auth and RBAC, enterprise deployment patterns.

Sample Outputs

Domain	Days Left	Status	Priority
google.com	62	HEALTHY	Low
wikipedia.org	27	WARNING	High
openai.com	-1	BROKEN	Critical

The table above represents the ranked output used for alerting, dashboard display and exports.

SSL Ops Intelligence Dashboard

10

TOTAL DOMAINS

8

HEALTHY

1

WARNING

1

BROKEN / ERROR

All Statuses ▾

Apply Filters

Download CSV

Download Markdown

DOMAIN	EXPIRY DATE	DAYS LEFT	STATUS	INCIDENT	RENEWAL STATUS	AI TASK (LLM INTELLIGENCE PLAN)	OWNER	CHECKED AT
openai.com	—	-1	BROKEN	BROKEN	NO_ACTION	📄 Task: Remediate Broken Domain openai.com 🔥 Priority: HIGH ⚙️ Action Steps: Audit domain configuration to identify potential causes of the BROKEN status for openai.com	secops-triage-team	2026-06-09 14:19
wikipedia.org	2026-07-06 20:52:29	27	WARNING	P2 - WARNING	RENEWAL_TRIGGERED	📄 Task: wikipedia.org Domain Expiration Warning Remediation 🔥 Priority: HIGH	secops-triage-team	2026-06-09 14:18